

ZX Cheat Card

For the CWI QEC poster session · Wed 26 Aug 16:30–18:00 · the diagrammatic bridge, in your pocket

The three terms (one line each)

Spiders — the atom. Green = Z-basis copy/merge, red = X-basis. Graph-state convention: green spider (0 in, 1 out) = $|+\rangle$; yellow H-box (2 in, 2 out) = CZ = the edge. ([van de Wetering, arXiv:2012.13966](#))

Pauli webs — the stabilizers, read off the diagram. A connected sub-diagram that evaluates to a Pauli operator; the code's stabilizers and logical operators are paths through the spider web. ([Wan–Price–Yao, arXiv:2601.04467](#))

Gadgets — the field's cost unit. Phase/Pauli gadgets $e^{i\alpha P}$; T-count optimization is gadgetization. The algebraic cousin of an energy cost — the seam this poster cares about. ([arXiv:1902.03178](#); [Pauli Gadget Synthesis, QPL 2026](#))

The one-line statement: "I represent the ultrametric geometry as a ZX graph state — the same language as the holographic-code construction on tree-like tessellations (Wan–Price–Yao 2026). The stabilizers are the Pauli webs of the diagram; the hierarchy is the tree."

The honest seam (say it when they engage): "It's a map, not the territory — but it's the same map the field draws." The diagrams import structure from different silos — internally consistent, externally unprobed — and the poster caption says so. That honesty is the legitimacy move (IAPS §6.2, audit the map).

Two questions that land

Q1 — the Bruhat–Tits bridge (yours, unique): "Does the resource state have to be a two-dimensional cluster — or can it be any graph, like a tree?" (tree-like resource states $\leftrightarrow$ MBQC; ties your T_2 graph state to the room's cluster-state machinery)

Q2 — the energy one (JPCUB): "If optimizing is that hard — has anyone tried optimizing for energy instead of gate count?" (T-count is the field's cost unit; joules is the one nobody prices)

Standing rules: ask about method, not math · never apologize for asking · exit line: "*can I follow up by email?*" · fig4 verification in hand: 22 spiders = 22 vertices, 21 H-boxes = 21 edges, $p=2 \Rightarrow$ 3-regular — same topology as fig2.